M:

General solution:

$$\Omega(E) = \frac{1}{\sqrt{2E}} \int \delta(R - \sqrt{2E})\, dV_{2N}$$

$$A = \frac{dV}{dr} \Longrightarrow dV = A\,dr$$

$$\Omega(E) = \frac{1}{\sqrt{2E}} \int \delta(R - \sqrt{2E})\, A_{2N-1}\, dR$$

Area element of 2N-Dimensional sphere is

$$A_{2N-1} = \frac{2\pi^N}{\Gamma(N)} R^{2N-1}$$

[Source in latex bibliography]

$\Gamma(N)$ is the gamma function, as N here is a positive integer

$$\Gamma(N) = (N-1)!$$

$$A_{2N-1} = \frac{2\pi^N}{(N-1)!} R^{2N-1}$$

$$\therefore \Omega(E) = \frac{1}{\sqrt{2E}} \frac{2\pi^N}{(N-1)!} \int \delta(R - \sqrt{2E}) R^{2N-1}\, dR$$

$$= \frac{1}{\sqrt{2E}} \frac{2\pi^N}{(N-1)!} (\sqrt{2E})^{2N-1}$$

$$= \frac{2\pi^N}{(N-1)!} (\sqrt{2E})^{2(N-1)} = \frac{2\pi^N}{(N-1)!} (2E)^N$$

$$\Omega(E) = \frac{2\, 2^{N-1} \pi^N E^{N-1}}{(N-1)!}$$

$$= \frac{2^N \pi^N}{(N-1)!} E^{N-1}$$

$$= \frac{(2\pi)^N}{(N-1)!} E^{N-1}$$

$$\therefore \Omega(E) = \frac{(2\pi)^N}{(N-1)!} E^{N-1}$$

$$\Rightarrow S = (N-1)\ln E + N\ln 2\pi - \ln(N-1)$$

$$S = (N-1)\ln E + N\ln 2 + N\ln\pi - \ln(N-1)!$$